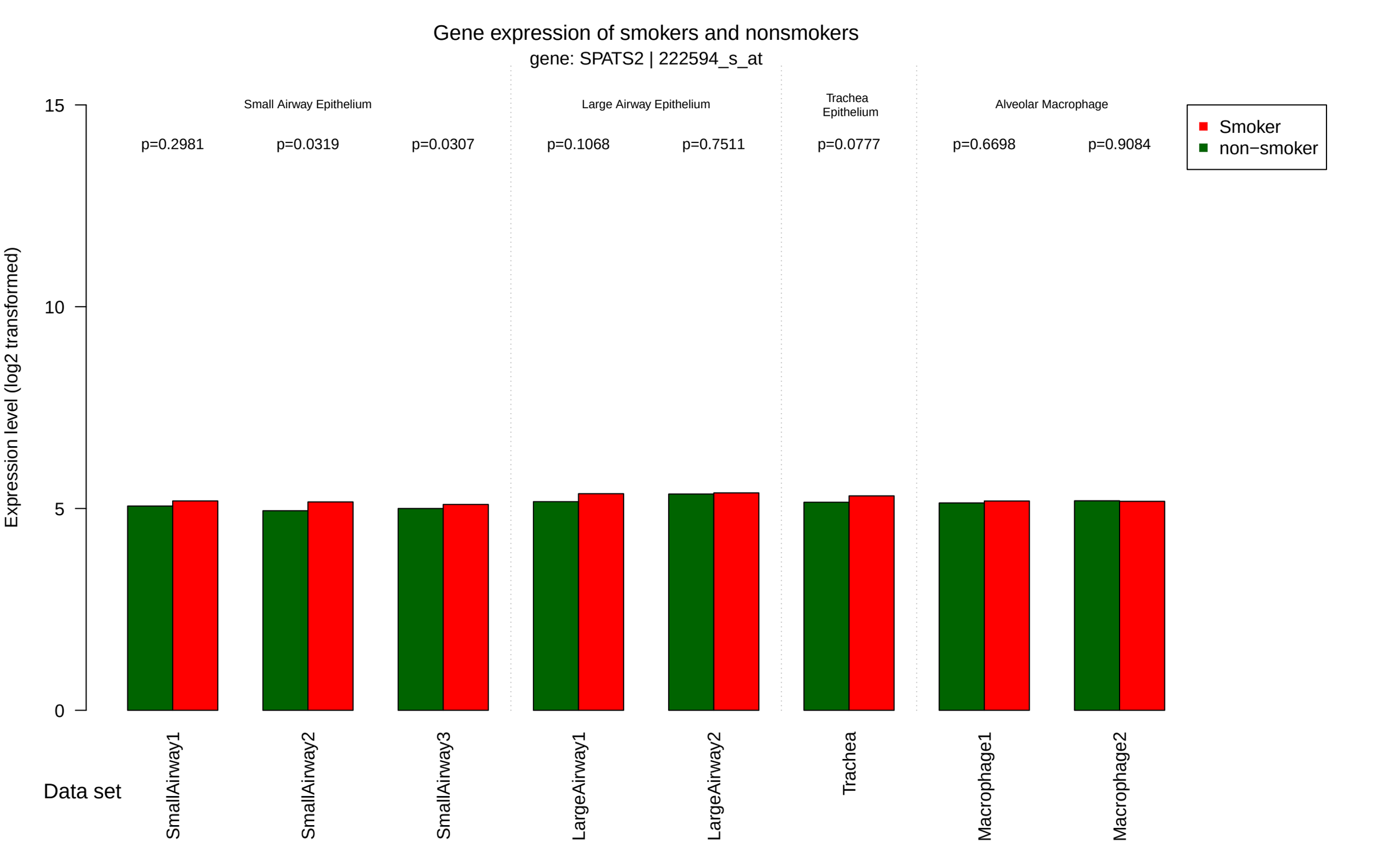
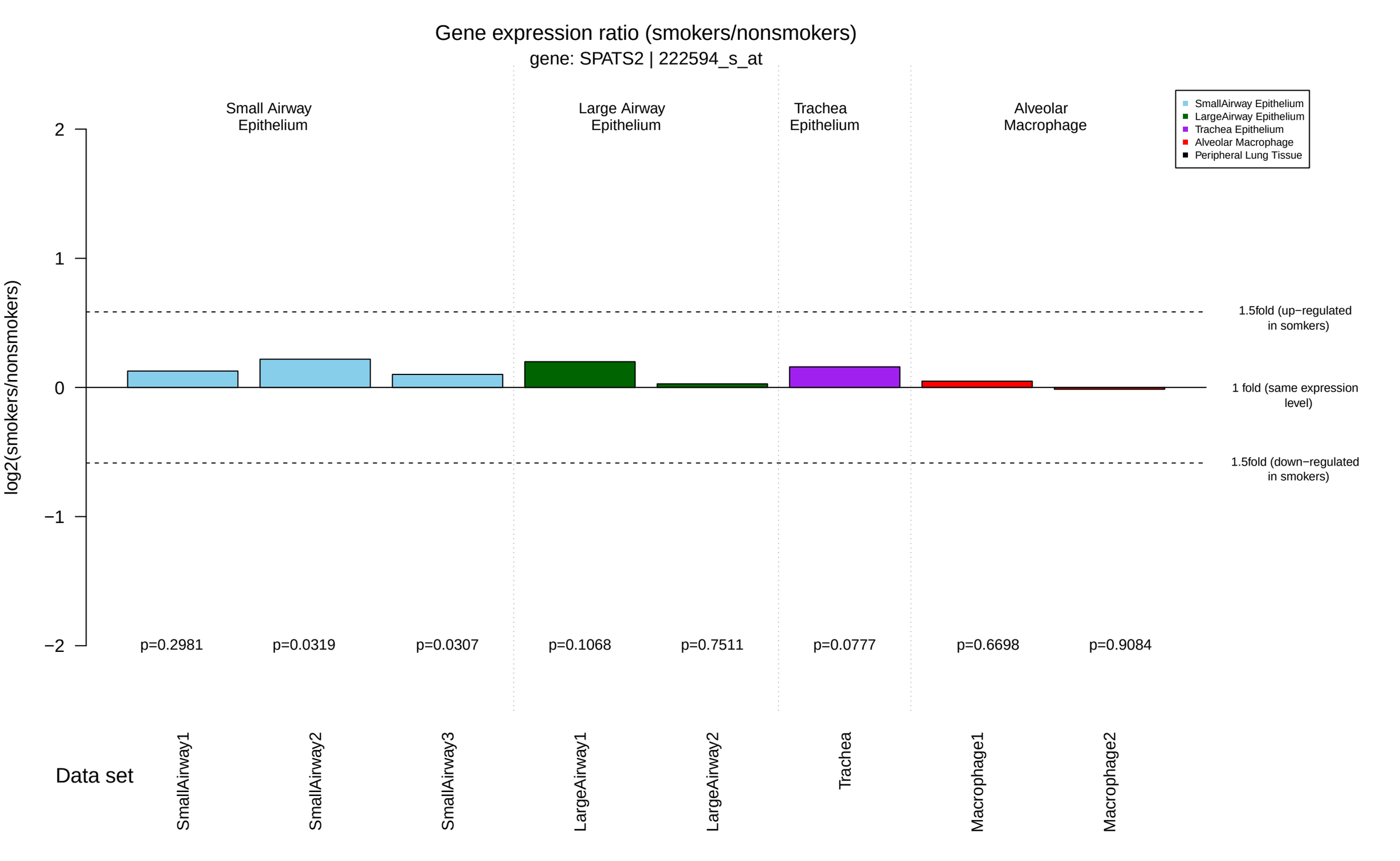
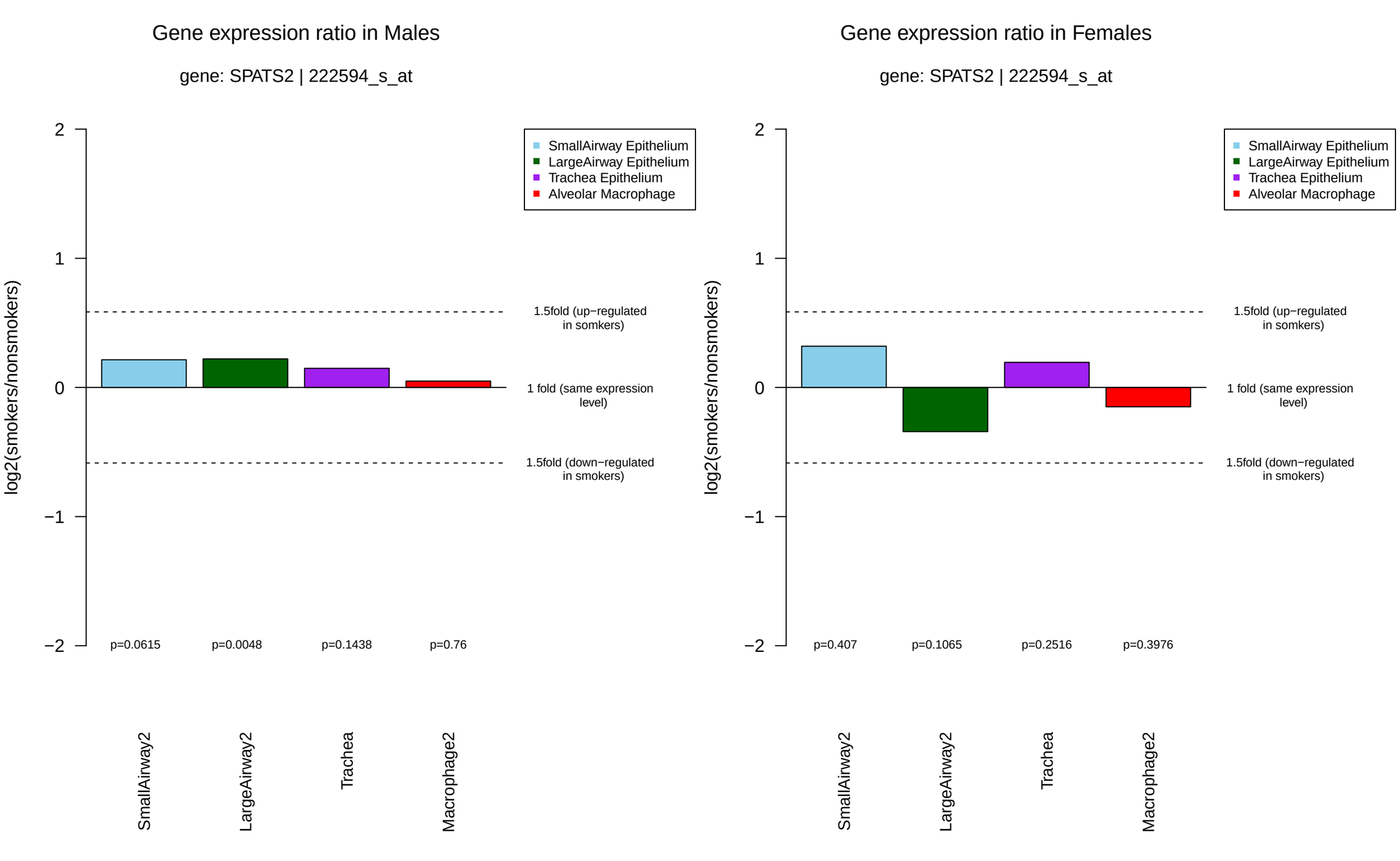
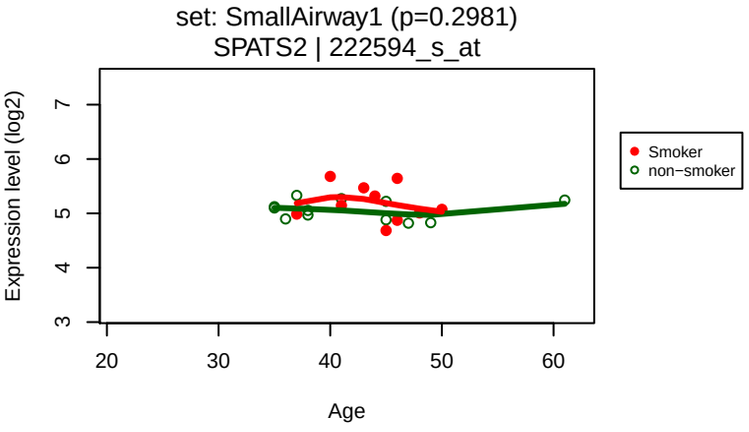
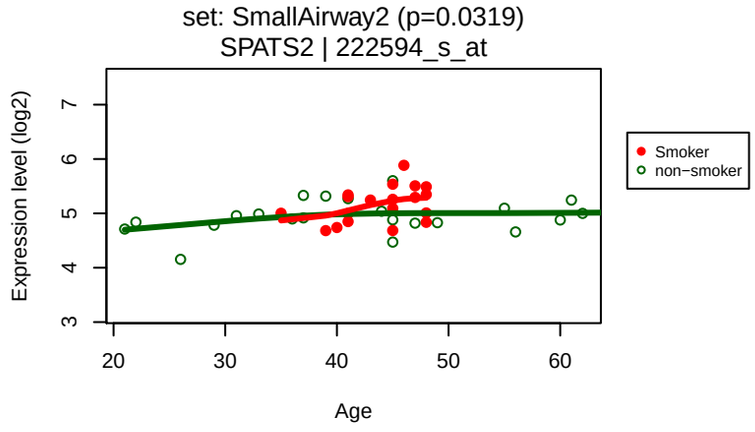
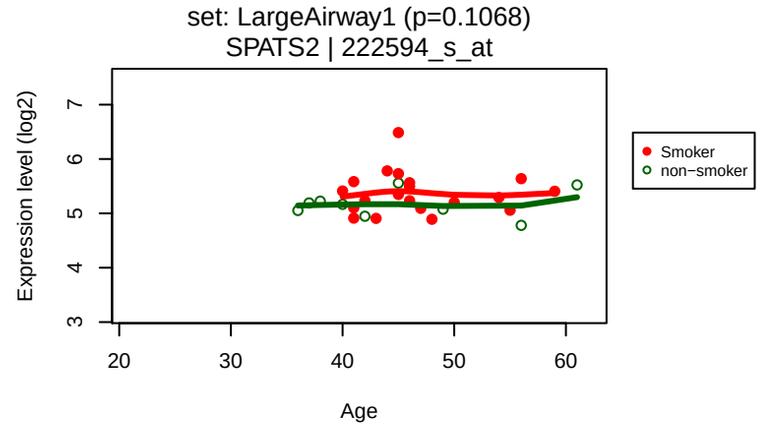

Gene expression of smokers and nonsmokers
gene: SPATS2 | 222594_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2981
p=0.0319
p=0.0307
p=0.1068
p=0.7511
p=0.0777
p=0.6698
p=0.9084
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

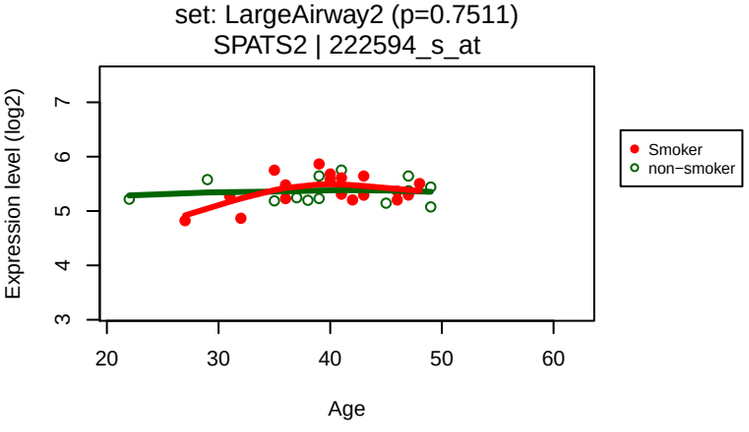

Gene expression ratio (smokers/nonsmokers)
gene: SPATS2 | 222594_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2981
p=0.0319
p=0.0307
p=0.1068
p=0.7511
p=0.0777
p=0.6698
p=0.9084
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

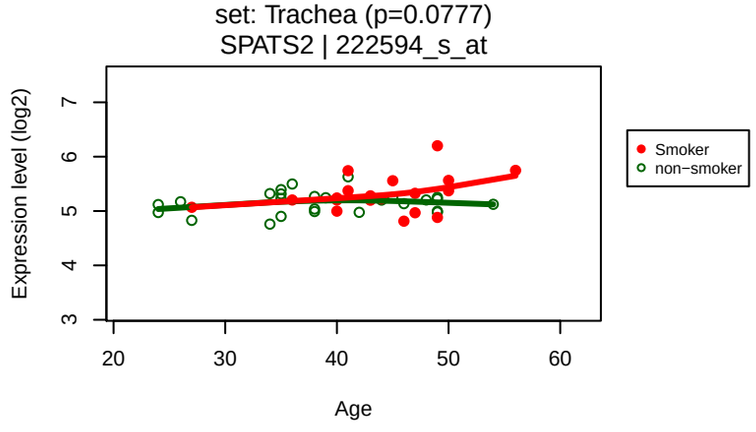

Gene expression ratio in Males
gene: SPATS2 | 222594_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0615
p=0.0048
p=0.1438
p=0.76
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATS2 | 222594_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.407
p=0.1065
p=0.2516
p=0.3976
SmallAirway2
LargeAirway2
Trachea
Macrophage2

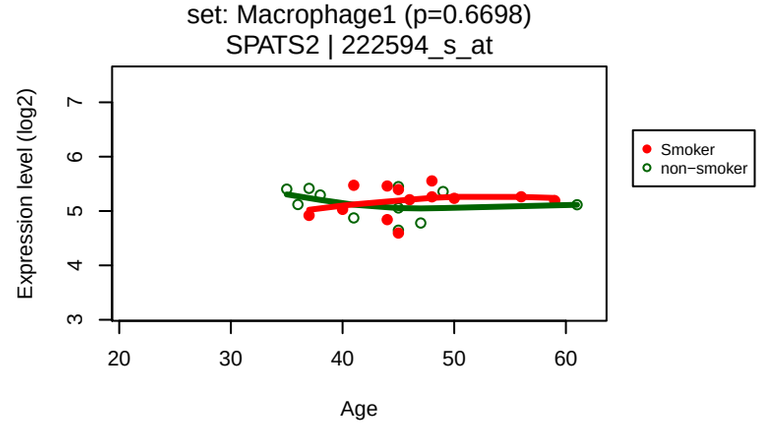

set: SmallAirway1 (p=0.2981)
SPATS2 | 222594_s_at
Expression level (log2)
Age
Smoker
non−smoker

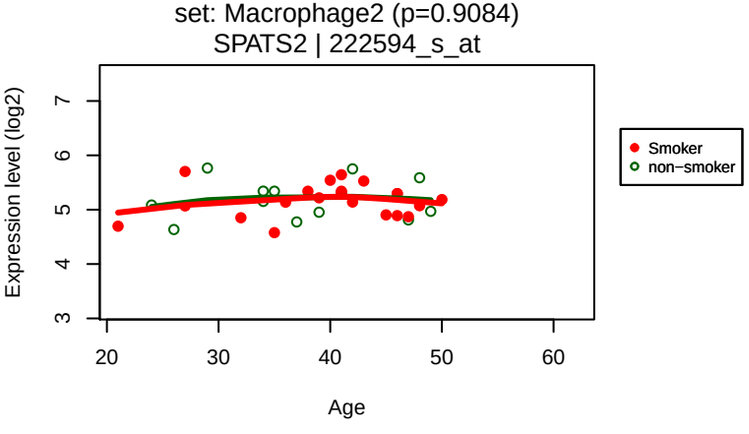

set: SmallAirway2 (p=0.0319)
SPATS2 | 222594_s_at
Expression level (log2)
Age
Smoker
non−smoker

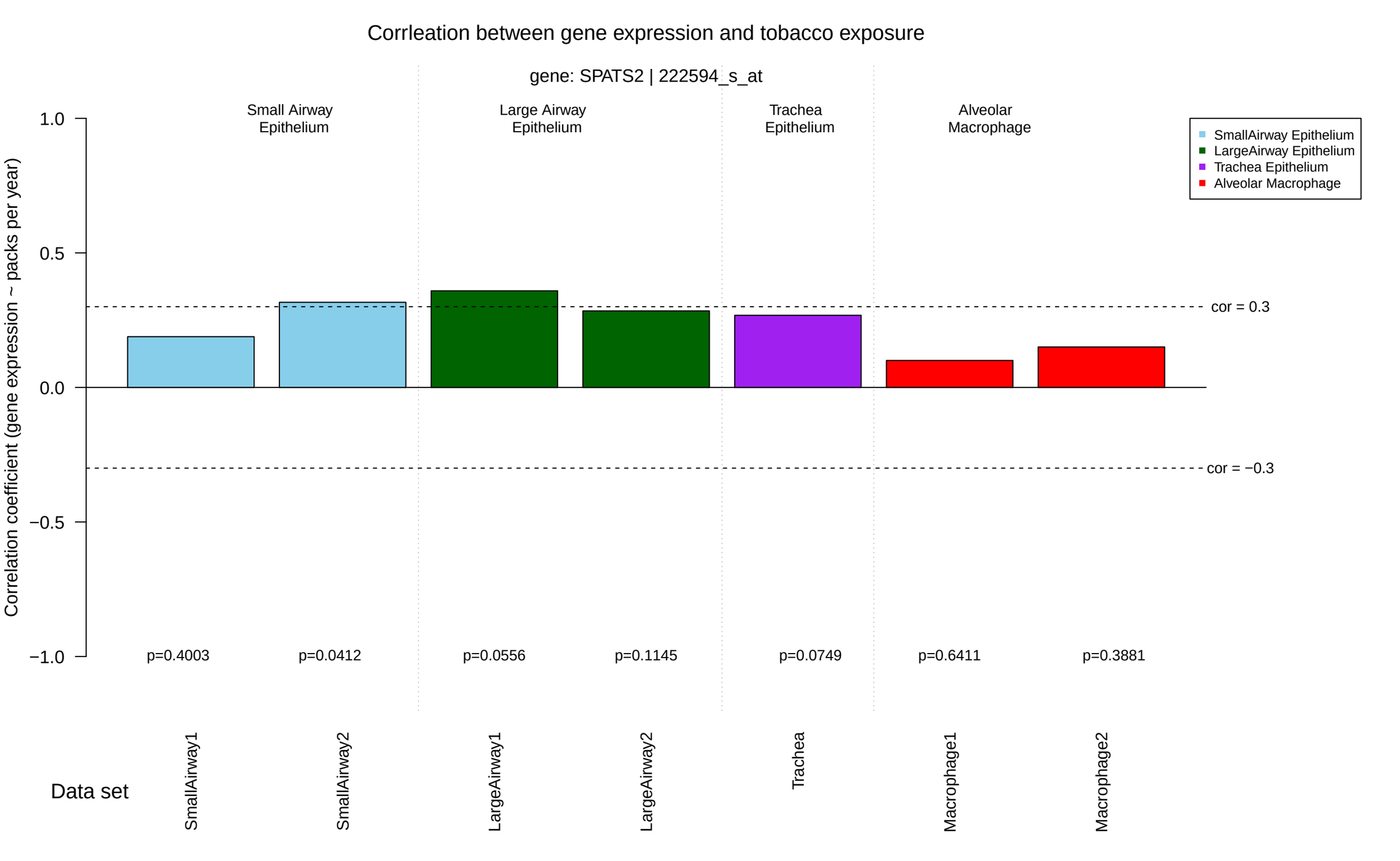

set: LargeAirway1 (p=0.1068)
SPATS2 | 222594_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7511)
SPATS2 | 222594_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0777)
SPATS2 | 222594_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6698)
SPATS2 | 222594_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9084)
SPATS2 | 222594_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATS2 | 222594_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4003
p=0.0412
p=0.0556
p=0.1145
p=0.0749
p=0.6411
p=0.3881
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2